

FFIMS DEVELOPMENT REPORT

A1 Fleet Management Module

Post-Presentation Development Phase

April 22, 2026

Prepared for: Ms. Mafu

Project Supervisor

Team Composition

Group A1 - Fleet Management Team:

No.	Student Name	Role/Contribution
1	NYEMBO JENOVIC MWEHU	Team Lead / Backend Development
2	Michael Tavonga MAFEMBA	Frontend Development
3	Joseph MALUNGA	Database Design
4	Mathew Wycklife ODHIAMBO	API Integration
5	Silva De Jesus Eduardo Pinto Carvalho	Testing & Documentation

Module Scope & Features

The A1 Fleet Management module is responsible for comprehensive fleet operations covering:

Core Features

1. Vehicle Management

- Vehicle registration and profiles
- Fleet inventory tracking
- Vehicle status monitoring (Available, Assigned, Maintenance, Inactive)
- Vehicle specifications and documentation

2. Trip Management

- Trip planning and scheduling
- Trip authorization workflows
- Route optimization
- Trip history and logs

3. Fuel Management

- Fuel consumption tracking
- Fuel card management
- Refueling logs and receipts
- Fuel efficiency analytics
- Cost analysis per vehicle

4. Servicing & Maintenance

- Scheduled maintenance tracking
- Service history records

- Preventive maintenance alerts
- Workshop management
- Parts inventory integration

5. Driver Management

- Driver profiles and licensing
- Driver assignment to vehicles
- Driver performance metrics
- License expiry tracking

6. Mileage Tracking

- Odometer readings
- Distance analytics
- Mileage-based maintenance scheduling
- Top 6 vehicles by mileage reporting

7. Vehicle Availability

- Real-time availability status
- Booking system
- Conflict resolution
- Availability forecasting

⚠ CRITICAL INTEGRATION POINT

Fleet Management ↔ Shift & Workforce Scheduling

CRITICAL QUESTION: Has the A1 Fleet Management team formally discussed integration requirements with the Shift & Workforce Scheduling (SWS) module team?

Why This Integration Matters

The Fleet Management and Shift & Workforce Scheduling modules have **significant operational overlap** and **shared data dependencies**:

Shared Resource	Fleet Management Needs	SWS Module Needs
Drivers	Driver profiles, licenses, vehicle assignments	Driver availability, shift assignments, skill matching
Vehicles	Vehicle status, availability, maintenance	Vehicle allocation for shifts, compliance checks
Trips/Journeys	Trip planning, fuel tracking	Journey scheduling, driver rostering, handovers
Availability	Real-time vehicle status	Workforce-vehicle matching, standby activation
Compliance	Vehicle licensing, insurance	Labor Act compliance, rest periods, max driving hours

Immediate Action Items

For A1 Fleet Management Team:

1. Schedule Integration Meeting (URGENT)

Meet with SWS module team (Nature & team)

Duration: 2 hours minimum

Agenda: Review overlapping features, define data ownership, agree on integration approach, design shared API contracts

2. Document Shared Entities

Create comprehensive data dictionary for: Drivers, Vehicles, Trips/Journeys

Include fields, validation rules, and constraints

Share with SWS team for review

3. Define API Contracts

Fleet → SWS APIs:

GET /api/vehicles/{id}/availability

GET /api/drivers/{id}/status

GET /api/trips/active

SWS → Fleet APIs:

POST /api/drivers/{id}/assign-shift

PUT /api/vehicles/{id}/reserve

GET /api/drivers/available

Integration Timeline Proposal

Week	Activity	Responsible Party
Week 1	Integration kickoff meeting	Both teams
Week 2	Shared schema design	Database leads (A1 + SWS)
Week 3	API contract definition	Backend leads (A1 + SWS)
Week 4	Implementation of shared endpoints	Both teams
Week 5	Integration testing	Testing leads (A1 + SWS)
Week 6	Bug fixes and optimization	Both teams

Recommendations for Ms. Mafu

1. Mandate Cross-Module Integration Meetings

- Require all teams with overlapping domains to meet weekly
- Document integration decisions in shared repository
- Review integration architecture in next presentation

2. Establish FFIMS Integration Committee

- Representatives from A1 (Fleet), P5 (SWS), and other overlapping modules
- Regular meetings to resolve integration issues
- Authority to make architectural decisions

3. Create Shared Data Dictionary

- FFIMS-wide entity definitions
- Prevent duplicate implementations
- Ensure consistency across modules

Conclusion

The A1 Fleet Management module has made solid progress in core features. However, **critical integration with the Shift & Workforce Scheduling module has not been adequately addressed**. Given the significant overlap in functionality (drivers, vehicles, trips), failure to coordinate will result in:

- Data duplication and inconsistencies
- User confusion (competing interfaces)
- Integration nightmares during final system assembly
- Potential project delays

Immediate action is required to establish formal integration protocols with the SWS team. This report recommends prioritizing cross-module collaboration in the next sprint to ensure seamless FFIMS operation.

Prepared by: NYEMBO JENOVIC MWEHU (A1 Team Lead)

Reviewed by: A1 Team Members

Submitted to: Ms. Mafu (Project Supervisor)

Date: April 22, 2026

END OF REPORT